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Sexually Transmitted Diseases

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Keywords

anal and perianal ulcers; anogenital ulcers; ectoparasitic infections; epidemiology of sexually transmitted diseases; epididymitis; genital; human immunodeficiency virus/acquired immunodeficiency syndrome and the urologist; urethritis; vaginitis; zika virus

Questions

1. The most commonly diagnosed bacterial sexually transmitted infection (STI) in the United States is:
    a. gonorrhea.
    b. ureaplasma.
    c. syphilis.
    d. chlamydia.
    e. chancroid.
2. The major health risk to untreated *Chlamydia* infection in men is:
    a. epididymitis.
    b. Reiter syndrome.
    c. orchitis.
    d. chronic prostatitis/chronic pelvic pain syndrome.
    e. transmission to a female partner resulting in pelvic inflammatory disease.

3. In addition to treatment for chlamydia, what other medication is recommended as a first-line treatment for gonorrhea?
    a. Ciprofloxacin
    b. Levofloxacin
    c. Ceftriaxone
    d. Cefixime
    e. Penicillin VK
4. Which subtypes of human papillomavirus (HPV) are responsible for development of malignancies including cervical, penile, and anal?
    a. 16 and 18
    b. 13 and 14
    c. 6 and 11
    d. 31 and 33
    e. 26 and 28
5. HPV vaccines are indicated for which groups?
    a. All sexually active women
    b. All sexually active men who have sex with men (MSMs)
    c. Men and women up to age 26 years
    d. Only women up to age 26 years
    e. Only women with a family history of cervical cancer
6. Which STI often has no visible genital lesion because it has usually resolved by the time of presentation and is associated with tender, often suppurative adenopathy?
    a. Chancroid
    b. Herpes simplex virus type 2 (HSV-2)
    c. Herpes simplex virus type 1 (HSV-1)
    d. Donovanosis
    e. Lymphogranuloma venereum (LGV)
7. Which of the following is not a reportable STI in every state?
    a. HSV
    b. Syphilis
    c. Chancroid
    d. Chlamydia
    e. Human immunodeficiency virus (HIV)
8. Which of the following tests should be used to monitor the clinical response to treatment in patients with syphilis?

a. *Treponema pallidum* particle agglutination (TP-PA)
b. Rapid plasma reagin (RPR)
c. Fluorescent treponemal antibody absorption (FTA-ABS)
d. Darkfield microscopy
e. Tzanck test

9. What is the treatment of choice for primary, secondary, and early latent syphilis?
   a. Azithromycin
   b. Benzathine penicillin
   c. Probenecid penicillin
   d. Ceftriaxone
   e. Procaine penicillin
10. Which statement is true regarding the likelihood of recurrent genital lesions in patients with HSV-1 and HSV-2?
   a. No difference except in HIV patients in whom HSV-1 recurs more often
   b. No difference except in HIV patients in whom HSV-2 recurs more often
   c. HSV-1 recurs more often
   d. HSV-2 recurs more often
   e. No difference
11. What is the causative organism for LGV?
   a. *Calymmatobacterium*
   b. *Klebsiella*
   c. *Chlamydia*
   d. *Haemophilus ducreyi*
   e. *Trichomonas*
12. Gardasil in males is recommended:
   a. to prevent transmission of HPV to their partners.
   b. only for persons with documented HPV.
   c. to prevent anal cancer and genital warts.
   d. to prevent genital warts only.
   e. only in men older than 26 years.
13. Donovan bodies are seen on:
   a. Thayer-Martin medium in gonorrhea.
   b. ulcer scraping in primary syphilis.

c. biopsy of condyloma acuminata.
d. lymph node aspiration in LGV.
e. biopsy of ulcer in granuloma inguinale.

14. Clue cells are diagnostic of:
    a. bacterial vaginosis (BV).
    b. *Trichomonas.*
    c. scabies.
    d. *Candida* vulvovaginitis.
    e. *Torulopsis glabrata* vulvovaginitis.
15. HIV is what type of virus?
    a. Single-stranded RNA
    b. Double-stranded RNA
    c. Single-stranded messenger RNA (mRNA)
    d. Single-stranded DNA
    e. Double-stranded DNA
16. The HIV envelope precursor protein gp160 is cleaved into what two envelope proteins?
    a. gp123 and gp33
    b. gp124 and gp40
    c. gp141 and gp20
    d. gp120 and gp41
    e. gp120 and gp6
17. HIV infects which immune cells?
    a. Macrophages
    b. B cells
    c. CD4 T cells
    d. CD8 T cells
    e. Natural killer (NK) cells
18. The initial screening test for HIV is:
    a. Western blot.
    b. indirect immunofluorescence assay.
    c. nucleoside amplification assay.
    d. rapid enzyme immunoassay (EIA).
    e. viral culture.
19. Antiretroviral therapy is recommended for:
    a. patients with CD4 count of <200 only.

b. patients with CD4 count of <350 only.
c. patients with CD4 count of <500 only.
d. patients with CD4 count of <500 and symptoms.
e. all patients with HIV.

20. What factor is not associated with increased risk of seroconversion with HIV after a needle stick?
    a. Recent infection by patient
    b. Deep exposure of needle
    c. Visible blood on injuring device
    d. Prior placement of injuring device in artery or vein
    e. Patient dying within 2 months of exposure
21. Polymorphisms of which gene are associated with development of HIV-associated nephropathy (HIVAN) in African American patients?
    a. Tyrosine kinase
    b. Antichymotrypsin-1
    c. Apolipoprotein-1
    d. Tumor necrosis alpha
    e. Interleukin-10
22. Which type of antiretroviral medications are associated with formation of urinary tract stones?
    a. Integrase inhibitors
    b. Fusion inhibitors
    c. Protease inhibitors
    d. Non-nucleoside reverse-transcriptase inhibitors (NNRTIs)
    e. CCR5 blockers
23. What is the best diagnostic test to detect HIV in the acute phase of infection?
    a. Indirect immunofluorescence assay
    b. Viral load assay
    c. Western blot
    d. HIV-1/HIV-2 serology assay
    e. Rapid EIA
24. What class of medication can have a prolonged half-life when used in association with protease inhibitors and NNRTI's for treatment of HIV?

a. Alpha-blockers
b. 5-Alpha reductase inhibitors
c. Beta-3 agonists
d. PDE5 inhibitors
e. Fluoroquinolones

25. What type of genitourinary (GU) cancer is not increased in frequency in patients with HIV?
    a. Prostate cancer
    b. Kidney cancer
    c. Penile cancer
    d. Testis cancer
    e. Kaposi sarcoma

Pathology

1. A 32-year-old sexually active woman has the lesion in Fig. 58.1A and B excised from her vulva. The diagnosis is condyloma acuminata. The most appropriate next step is:
    a. acetic acid test.
    b. podophyllin to the base of the lesion.
    c. HPV vaccine.
    d. vaginoscopy.
    e. cystoscopy.
2. A 22-year-old sexually active man has a 2-mm raised red lesion in the suprapubic area. A biopsy is performed (depicted in Fig. 58.2) and read as molluscum contagiosum. The patient is concerned and desires treatment. The most appropriate treatment is:
    a. wide local excision.
    b. 5-fluorouracil cream.
    c. ciprofloxacin for 2 weeks.
    d. bactrim for 4 weeks.
    e. liquid nitrogen application.

Answers

1. **d. Chlamydia. Chlamydia is the most common bacterial STI in the United States.** The 1,422,976 cases of *Chlamydia trachomatis* infection reported to the Centers for Disease Control and Prevention (CDC) in 2012 was the largest number ever reported to the CDC for any condition. The second most common bacterial STI is gonorrhea.

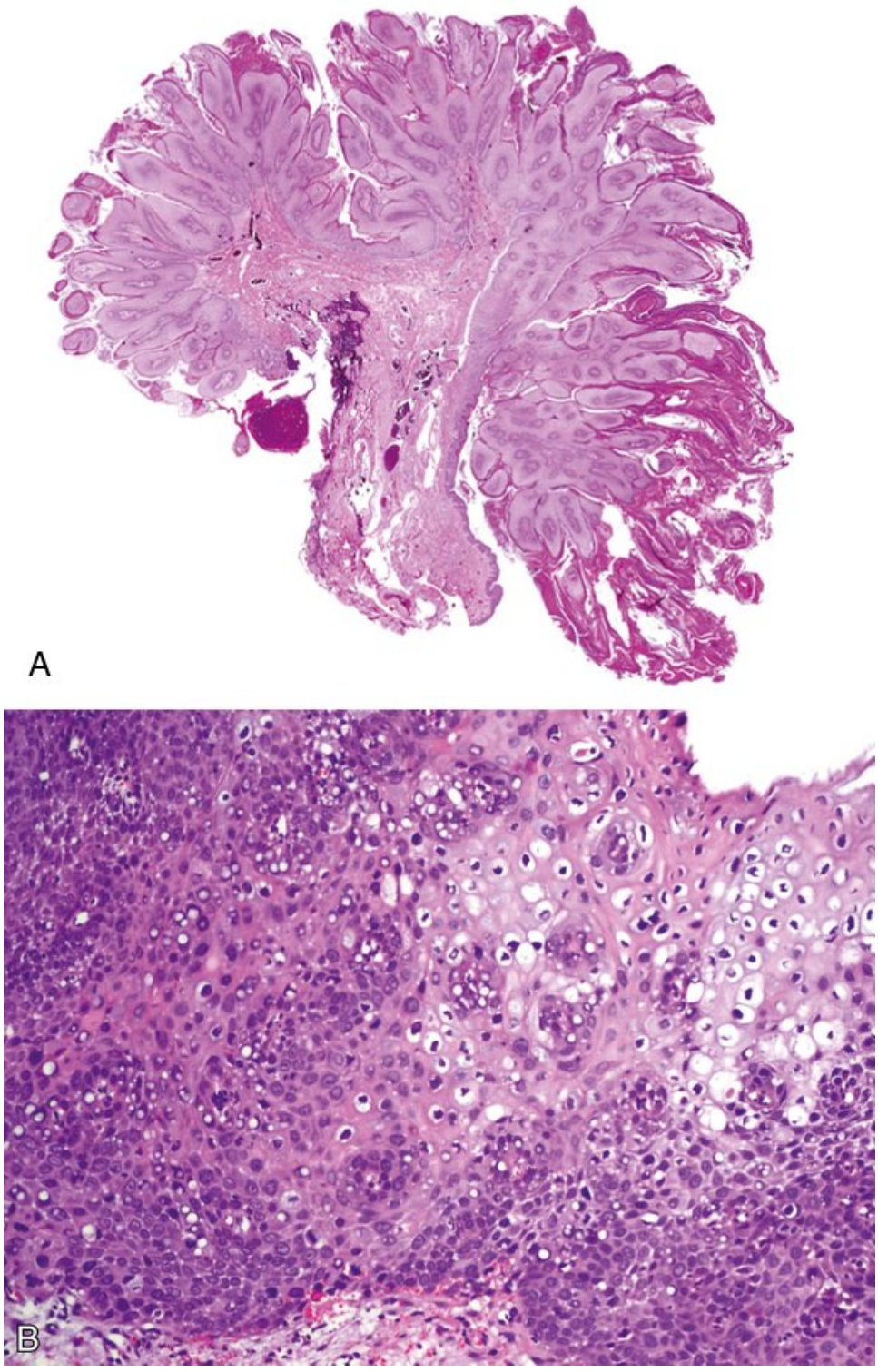


FIG. 58.1 A and B.
From Bostwick DG, Cheng L: *Urologic surgical pathology*, ed 3, St. Louis, 2014, Saunders.

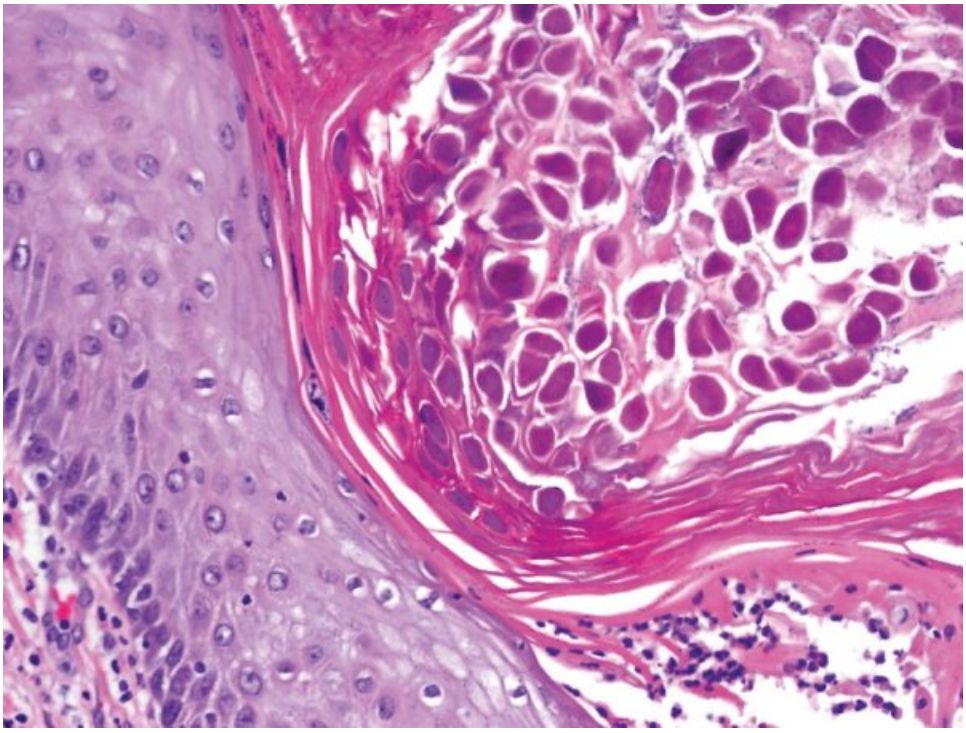


FIG. 58.2
From Bostwick DG, Cheng L: *Urologic surgical pathology*, ed 3, St. Louis, 2014, Saunders.

2. **e. Transmission to a female partner resulting in pelvic inflammatory disease.** Up to 75% of women with chlamydial infection can be asymptomatic. Ascending chlamydial infection can result in scarring of the fallopian tubes, pelvic inflammatory disease, risk for ectopic pregnancy, pelvic pain, and infertility. The **risk of untreated chlamydial infection producing pelvic inflammatory disease is estimated to be between 9.5% and 27% of cases.**
3. **c. Ceftriaxone.** As of 2007, quinolones are no longer recommended in the United States for treatment of gonorrhea and associated conditions such as pelvic inflammatory disease. As of August 2012, because of high resistance, cefixime is no longer recommended as first-line therapy to treat gonorrhea. **Current treatment of uncomplicated gonococcal infections of the cervix, urethra, and rectum is ceftriaxone, 250 mg IM, single-dose PLUS azithromycin, 1 g orally in single dose.**

4. **a. 16 and 18.** Types 6 and 11 are nononcogenic and are responsible for about 90% of anogenital warts. Other subtypes, including 16 and 18, account for cervical cancer and other types of anogenital cancer including vulvar, vaginal, anal, and penile.
5. **c. Men and women up to age 26 years.** In June 2006, a quadrivalent HPV vaccine (Gardasil; Merck, Sharpe, and Dohme Corporation) was licensed for use in the United States in females aged 9 to 26 years. In October 2009, this vaccine also was licensed for use in males aged 9 to 26 years. This vaccine provides protection against HPV types 6, 11, 16, and 18. In October 2009, a bivalent HPV vaccine (Cervarix; Glaxo SmithKline Biologicals, Research Triangle Park, NC) that provides protection against types 16 and 18 was licensed for use in females aged 10 to 25 years.
6. **e. Lymphogranuloma venereum (LGV).** A self-limited genital ulcer or papule sometimes is present at the site of infection but usually has disappeared by the time of presentation. The secondary stage is the most common presentation in heterosexuals and is **marked by tender inguinal and/or femoral lymphadenopathy, typically unilateral.**
7. **a. HSV.** Diseases that must be reported to local health authorities: Syphilis, gonorrhea, chlamydia, chancroid, HIV infection, and acquired immunodeficiency syndrome (AIDS) are reportable diseases in every state.
8. **b. Rapid plasma reagin (RPR). Nontreponemal tests (RPR or VDRL) are used to monitor disease activity.** A fourfold change in titer equivalent to a change of two dilutions (e.g., from 1:16 to 1:4) is considered necessary to demonstrate a clinically significant difference.
9. **b. Benzathine penicillin.** Benzathine penicillin is the treatment of choice for all of the stages of syphilis. Treatment varies by dose and duration of therapy. Not considered appropriate treatments are combinations of benzathine and procaine penicillin (Bicillin C-R), or oral penicillin.
10. **d. HSV-2 recurs more often.** Genital HSV-1 recurs much less frequently (0.02 per month) than genital HSV-2 infections (0.23 per month), on the order of 10-fold less.

11. **c. *Chlamydia.*** LGV is an infection by *Chlamydia*, specifically serovars L1, L2, or L3.
12. **c. To prevent anal cancer and genital warts.** The quadrivalent vaccine is used in males to prevent genital warts and in both genders to prevent anal cancer. MSMs are particularly at risk for developing anal intraepithelial neoplasia and anal cancer. As in women, it is best started in males before the onset of sexual activity.
13. **e. Biopsy of ulcer in granuloma inguinale.** Granuloma inguinale (GI) is an infection by the intracellular gram-negative bacterium *Klebsiella granulomatis* (formerly called *Calymmatobacterium granulomatis*) that produces genital ulcers. The bacterium is a strict human pathogen, which makes culture difficult. **Diagnosis requires visualization of dark-staining Donovan bodies on crush preparation or biopsy, described by Donovan in 1905. These are intracellular inclusions of the bacteria within the cytoplasm of macrophages and appear deep purple when stained with a Wright, Giemsa, or Leishman stain.**
14. **a. Bacterial vaginosis (BV). Characteristic findings for BV on microscopic exam are clue cells, which are vaginal epithelial cells covered with bacteria.**
15. **a. Single-stranded RNA.** The genetic material in HIV is single-stranded RNA. After entry into the targeted cell, the RNA is reverse transcribed by a reverse transcriptase into a double-stranded DNA. This new DNA is assembled into complexes, which then associate with the target cell chromatin and integrate via the action of viral integrase. The cell then translates and transcribes the viral genes to produce proteins that will assemble new copies of the virus. Copies of the virus are called virions.
16. **d. gp120 and gp41.** The virus is shaped like a sphere. It is covered by an outer envelope that is a lipid bilayer derived from the host cell when it buds out of the cell. Embedded in the envelope is a complex of proteins known as Env. There is initially a precursor gp160 that is cleaved by a protease in the trans-Golgi network. It is cleaved into an outer subunit glycoprotein 120 (gp120) and a

transmembrane subunit glycoprotein 41 (gp41). After proteolysis, the gp120 and gp41 remain coupled as noncovalent heterodimers.

17. **c. CD4 T cells.** Glycoprotein gp120 has a high-affinity binding site for the T lymphocyte receptor CD4.
18. **d. Rapid enzyme immunoassay (EIA).** Diagnosis of HIV includes the use of serologic tests that detect antibodies against HIV-1 (and HIV-2) and by virologic tests that detect HIV antigens RNA. **The initial test is a screening test for antibodies, the conventional or rapid EIA. The initial result can be obtained in 30 minutes. Positive or reactive screening tests must be confirmed by a supplemental antibody test, Western blot, and indirect immunofluorescence assay (IFA) or by a virologic test, the HIV-1 RNA assay.**
19. **e. All patients with HIV.** The benefit of treatment may depend on the starting CD4 count, but treatment guidelines recommend treatment for all patients regardless of CD4 count.
20. **a. Recent infection by patient.** A review of **factors for increased risk of infection done by the CDC identified four that increased risk: deep as opposed to superficial exposure (odds ratio [OR] 15, 95% confidence index [CI] 6 to 41), visible blood on the injuring device (OR 6.2, 95% CI 2.2 to 21), prior placement of the injuring device in an artery or vein (OR 4.3, 95% CI 1.7 to 12), and patient dying within 2 months of the exposure (preterminal disease) (OR 5.6, 95% CI 2 to 16).**
21. **c. Apolipoprotein-1.** African Americans carrying two variants of the APOL-1 gene are at very high risk for HIVAN. These genes encode a secreted lipid binding protein called apolipoprotein-1 (apoL1). The variants G1 and G2 are common in African chromosomes but absent in European chromosomes; these variants lyse trypanosomes, including *Trypanosoma brucei rhodesiense*, which causes African sleeping sickness. Thus, these loci are thought to be selected out in this population. The presence of these two genes together increases the risk 29-fold, resulting in a 50% risk of developing HIVAN in untreated individuals as compared with a 12% baseline risk. Focal segmental glomerulosclerosis

(FSGS) found in individuals with the two risk genes also occurs at an earlier age and progresses much more rapidly.

22. **c. Protease inhibitors.** The protease inhibitors specifically have the possibility of stone formation. **Indinavir can form crystals in the urine. Indinavir stones are typically radiolucent on both plain film and computed tomography scan but can also be mixed with calcium and appear radio-opaque. Newer inhibitors including lopinavir, atazanavir, amprenavir, and nelfinavir have also been associated with the development of stones, but with less frequency than reported for indinavir.**
23. **b. Viral load assay.** During this initial 3-month period, the "window" period, antibody screening tests may be negative but the person is still infected. Virologic tests for HIV-1 RNA can be used to detect an acute infection in persons negative for HIV antibodies.
24. **d. PDE5 inhibitors.** PDE5 inhibitors depend on CYP3A for clearance, and all protease inhibitors and NNRTIs are inhibitors of CYP3A to some extent. **This can lead to a significant increase in the serum dose of PDE5 inhibitors, and therefore they should be started at the lowest dose possible in patients on these antiretroviral medications.**
25. **a. Prostate cancer.** The relative risk of prostate cancer in men with HIV compared to uninfected individuals has been reported as either no different or even less at 0.70.

Pathology

1. **d. Vaginoscopy.** The figure shows marked papillomatosis with koilocytic atypia. The patient should have a thorough genital examination, including vaginoscopy for other lesions.
2. **e. Liquid nitrogen application.** The biopsy shows an epidermal crater filled with molluscum bodies. This is usually a self-limited disease and requires no treatment. If treatment is desired, a local therapy is appropriate, such as curettage or liquid nitrogen application.

Chapter Review

1. When exposed to STIs, women are more likely to become infected and less likely to be symptomatic.
2. Chlamydia is the most common sexually transmitted disease in the United States.
3. HSV-2 accounts for 90% of the genital herpes infections. HSV-1 accounts for the remainder and is the common cause of cold sores; silent infection is common in this disease. The diagnosis is made by viral culture and subtyping. HSV enters the nerve and remains latent in the nerve cell body. It may cause aseptic meningitis and autonomic dysfunction, which may lead to urinary retention.
4. Chancroid is caused by *Haemophilus ducreyi* and results in a painful, nonindurated ulcer covered by an exudate. Inguinal adenopathy occurs and may become suppurative.
5. Chancre of syphilis is single, painless, indurated, and clean. It is associated with nontender inguinal lymphadenopathy.
6. Latent syphilis is seropositive with no evidence of disease. Early latent syphilis occurs in less than 1 year. Late latent syphilis occurs beyond 1 year.
7. Primary syphilis is the acute infection. Secondary syphilis is manifested by mucocutaneous and constitutional signs and symptoms that are often associated with a maculopapular rash. Tertiary syphilis is a systemic disease involving the cardiovascular, skeletal, and central nervous system.
8. Treponemal tests for syphilis are generally positive for life and do not indicate treatment response. RPR, Venereal Disease Research Laboratory (VDRL), and the toluidine red unheated serum test (TRUST) are nontreponemal tests and correlate with disease activity. They usually become negative after treatment. Nontreponemal tests (RPR or VDRL) are used to monitor disease activity.
9. The Jarisch-Herxheimer reaction occurs when patients with syphilis are treated with penicillin, resulting in the release of toxic products

when the treponemes are killed. The symptoms include headache, myalgia, fever, tachycardia, and increased respiratory rate.

10. LGV presents as a single painless ulcer and painful inguinal adenopathy. LGV is marked by tender inguinal and/or femoral lymphadenopathy, typically unilateral.
11. Polymerase chain reaction (PCR) assays are used for diagnosing chlamydial infection.
12. A strawberry rash on the vulva or strawberry cervix is seen in trichomoniasis.
13. More than 99% of cervical cancers and 84% of anal cancers are associated with HPV 16 or 18. The most common serotype associated with squamous cell carcinoma of the penis is HPV 16.
14. Biopsies of genital warts are not routinely indicated but should be performed when the wart is atypical, pigmented, indurated, or fixed and ulcerated.
15. HPV vaccine is recommended for females age 9 to 26 years and may also be given to males of the same age range.
16. The presence of STIs increases the risk for concurrent HIV.
17. Ulcerative STIs including herpes, syphilis, and chancroid enhance the susceptibility to HIV per sexual contact.
18. Antiviral therapy for HIV does not necessarily make the patient noninfectious.
19. Men who are circumcised are at lower risk for HIV infection.
20. There are two types of HIV viruses: HIV-1 and HIV-2. There are very few cases of HIV-2 in the developed world, and it is less easily transmitted and less virulent than HIV-1.
21. HIV is a retrovirus that infects T cells and dendritic cells.
22. Antiretroviral combination therapy delays the rate of progression of the disease and prolongs survival.
23. Overt AIDS is marked by a low CD4+ T-cell count.
24. Plasma HIV RNA load is the most accurate predictor of disease progression.
25. The diagnosis of HIV is made by screening for anti-HIV-1 and anti-HIV-2 antibodies. If this is positive, confirmation is made by using Western blot analysis. After treatment, the nadir of plasma HIV RNA predicts long-term outcome.

26. HIV testing is recommended for anyone diagnosed with a STI or at risk for STIs.
27. HSV increases HIV replication in persons infected with both viruses.
28. HPV infection increases the risk for carcinoma, especially in HIV-infected hosts.
29. The most common intrascrotal pathologic process in AIDS patients is testicular atrophy.
30. Voiding dysfunction is common in patients with advanced HIV infection.
31. Urinary calculi have been associated with, most notably, protease inhibitors such as indinavir. These stones are soluble at an acidic pH.
32. HIV-associated nephropathy is a glomerular disease that often presents as proteinuria.
33. Patients with HIV are at particular increased risk for Kaposi sarcoma and non-Hodgkin lymphoma. Kaposi sarcoma presents as a raised, firm, indurated purplish plaque, reflecting the presence of abundant blood vessels, extravasated erythrocytes, and siderophages.
34. Human herpesvirus 8 is essential for all forms of Kaposi sarcoma.
35. HIV protease inhibitors are also potent antiangiogenic molecules and are useful in treating Kaposi sarcoma. However, localized lesions may be treated by irradiation, laser, cryotherapy, or intralesional injections of antineoplastic drugs. Corticosteroids should not be used to treat the lesions.
36. There is an increased incidence of Kaposi sarcoma, non-Hodgkin lymphoma, and cervical cancer in patients with HIV infection. There also appears to be an increased incidence of the following GU tumors: testicular, renal, and penile.
37. The risk of untreated chlamydial infection producing pelvic inflammatory disease is estimated to be between 9.5% and 27% of cases.
38. Current treatment of uncomplicated gonococcal infections of the cervix, urethra, and rectum are ceftriaxone 250 mg IM single-dose PLUS azithromycin, 1 g orally in single dose or doxycycline, 100 mg orally twice per day for 7 days.
39. Donovan bodies noted in granuloma inguinale are intracellular inclusions of the bacteria within the cytoplasm of macrophages and

appear deep purple when stained with a Wright, Giemsa, or Leishman stain.

40. Characteristic findings for bacterial vaginosis on microscopic exam are clue cells, which are vaginal epithelial cells covered with bacteria.
41. Factors that increase the risk for transmitting HIV include deep as opposed to superficial exposure (OR 15, 95% CI 6 to 41), visible blood on the injuring device (OR 6.2, 95% CI 2.2 to 21), prior placement of the injuring device in an artery or vein (OR 4.3, 95% CI 1.7 to 12), and patient dying within 2 months of the exposure (preterminal disease) (OR 5.6, 95% CI 2 to 16).
42. PDE5 inhibitors depend on CYP3A for clearance, and all protease inhibitors and NNRTIs are inhibitors of CYP3A to some extent. This can lead to a significant increase in the serum dose of PDE5 inhibitors, and therefore they should be started at the lowest possible dose in patients on these antiretroviral medications.

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Cutaneous Diseases of the External Genitalia

Richard Edward Link, and Nikki Tang

Keywords

allergic dermatitis; cutaneous dermatology; genitalia infections and infestations; psoriasis; skin conditions; skin lesions; skin neoplasms; ulcers

Questions

1. The periodic acid–Schiff stain is used to identify what organism in scraped or touched skin preparations?
    a. *Pseudomonas* sp.
    b. *Candida*
    c. *Corynebacterium minutissimum*
    d. Herpes simplex
    e. Molluscum contagiosum
2. Oral glucocorticosteroids are often used to treat dermatologic conditions and have a duration of effect lasting:
    a. 2 to 3 weeks.
    b. 30 to 90 minutes.
    c. 1 to 5 hours.
    d. 8 to 48 hours.
    e. 5 to 7 days.

3. The preferred dosage schedule for a short course of oral glucocorticosteroids used to treat a cutaneous disorder is:
    a. a single morning dose.
    b. a single evening dose.
    c. doses in the morning and evening.
    d. a dose every other day in the morning.
    e. redosing every 8 hours.
4. A 12-year-old boy has a long-standing history of asthma and occasional outbreaks of erythematous, pruritic papules on his scrotum and lower extremities. Which of the following options represents a rational approach to treating this condition?
    a. Long-term suppressive topical corticosteroids
    b. Frequent soaking in warm water to prevent the development of lesions
    c. Low-dose systemic corticosteroids
    d. The frequent application of emollients
    e. Application of a topical calcineurin inhibitor
5. Patch testing is a useful diagnostic test to identify:
    a. psoriasis.
    b. contact dermatitis.
    c. erythema gangrenosum.
    d. atopic dermatitis.
    e. Behçet disease.
6. The North American Contact Dermatitis Group identified a series of common allergens that were associated with contact dermatitis. Which allergen was the most common offending agent in contact dermatitis cases?
    a. Silver
    b. Textile dyes
    c. Ragweed
    d. Nickel sulfate
    e. Pet dander
7. A 35-year-old textile worker spills a small amount of green dye onto her left thigh. By the end of the workday, she is complaining of pain and burning over a 5-cm irregular patch of skin on her left thigh. What is the most likely diagnosis?

a. Erysipelas
b. Allergic contact dermatitis
c. Hailey-Hailey disease
d. Irritant contact dermatitis
e. Koebner phenomenon

8. Following a recent exacerbation of genital herpes, a 22-year-old man notes the development of erythematous papules and targetoid lesions on his thighs, scrotum, and oral mucosa. The best next course of action is:
   a. oral antihistamines.
   b. systemic corticosteroids.
   c. observation.
   d. oral acyclovir.
   e. topical corticosteroids.
9. A 19-year-old woman is 2 days into a course of sulfonamides for an *Escherichia coli* urinary tract infection. She develops painful labial erosions that progress to a generalized rash with the formation of blisters. The most likely diagnosis is:
   a. erythema multiforme minor.
   b. Reiter syndrome.
   c. Stevens-Johnson syndrome.
   d. pyoderma gangrenosum.
   e. Sézary syndrome.
10. A 42-year-old circumcised man has a history of widely distributed erythematous plaques—most severe on his knees, elbows, inguinal folds, and glans penis. The condition has waxed and waned since he was in his early 20s. What is an appropriate therapy during an exacerbation?
    a. Topical 3% liquor carbonis detergens in 1% hydrocortisone cream
    b. Oral psoralen combined with ultraviolet radiation (PUVA)
    c. Systemic corticosteroids
    d. Topical 5-fluorouracil cream
    e. Oral azathioprine
11. A 21-year-old man presents with dysuria, blurred vision, oral ulcers, and erythematous plaques in his genitalia. He has mild

soreness in his knees and ankles. He is negative for human immunodeficiency virus (HIV) and has no history of sexually transmitted disease. What is a likely risk factor for development of this disorder?

a. Genital herpes simplex
b. The human leukocyte antigen (HLA)-B27 haplotype
c. A history of atopic dermatitis
d. Exposure to benzene-containing chemicals
e. Family history of psoriasis

12. Which of the following statements is true about the treatment of symptomatic genital lichen planus?
    a. Systemic corticosteroids can prevent the development of lesions.
    b. In clinical trials, the most effective agent for treating lichen planus is systemic acitretin.
    c. Systemic corticosteroids can shorten the time to clearance of existing lesions from 29 to 18 weeks.
    d. Phytotherapy is the therapeutic modality of choice for treating lichen planus.
    e. Metronidazole is an effective and well-established first-line agent in the treatment of lichen planus.
13. The late stage of lichen sclerosus involving the glans penis is termed:
    a. keratinizing balanoposthitis.
    b. pseudoepitheliomatous, keratotic, and micaceous balanitis.
    c. bowenoid papulosis.
    d. balanitis xerotica obliterans.
    e. Hailey-Hailey disease.
14. Which of the following cutaneous conditions has been associated with an increased risk of squamous cell carcinoma?
    a. Lichen sclerosus et atrophicus
    b. Lichen planus
    c. Psoriasis
    d. Bullous pemphigoid
    e. Lichen nitidus

15. An 18-year-old man has a history of seizures following an automobile accident 2 weeks ago. He was sexually active before the accident. Today, he presents with a solitary, painful erosion on the penis. What course of action is appropriate at this time?
    a. Perform a urethral swab for gonorrhea and chlamydia
    b. Consult with neurology to alter his antiseizure medication regimen
    c. Start oral acyclovir
    d. Perform a punch biopsy of the lesion
    e. Start oral doxycycline
16. A 35-year-old, previously healthy woman has noted the rapid development of sharply demarcated, pruritic, red-brown plaques over a large extent of her skin surface. The plaques are particularly dense in her nasolabial folds and perianal area, and the nails are spared. What is the next step?
    a. Systemic corticosteroids
    b. HIV test
    c. Skin culture for *Malassezia furfur*
    d. Examination of the lesions under ultraviolet (UV) light
    e. Biopsy of the lesions
17. In patients with pemphigus vulgaris, the characteristic clinical sign showing loss of epidermal cohesion is:
    a. the dimple sign.
    b. the Asboe-Hansen sign.
    c. the Leser-Trélat sign.
    d. the dimple sign.
    e. the bullous blanching sign.
18. Which of the following statements is FALSE concerning pemphigus vulgaris?
    a. The majority of pemphigus patients have painful oral mucosal erosions.
    b. Pemphigus appears to have an autoimmune pathogenesis.
    c. Blisters appear to form due to loss of keratinocyte cell-cell adhesion.
    d. Treatment of pemphigus relies on systemic corticosteroids.

e. Given enough time, even advanced cases of pemphigus generally resolve spontaneously without sequelae.

19. Which of the following dermatoses has an association with celiac disease?
    a. Dermatitis herpetiformis
    b. Hailey-Hailey disease
    c. Behçet disease
    d. Bullous pemphigoid
    e. Psoriasis
20. Which of the following is not a vesicobullous dermatosis?
    a. Hailey-Hailey disease
    b. Pyoderma gangrenosum
    c. Pemphigus vulgaris
    d. Zoon balanitis
    e. Linear IgA bullous dermatoses
21. Which agent has been shown to be effective in treating linear IgA bullous dermatoses?
    a. Azathioprine
    b. Cyclosporine
    c. Dapsone
    d. Metronidazole
    e. Sulfonylurea
22. A 45-year-old woman has pruritic, foul-smelling blistering in the inframammary folds and groin. The skin findings are confluent areas of vesicles with fragile blisters. Which of the following statements is FALSE concerning this condition?
    a. The condition is usually worse during the summer months.
    b. Intralesional corticosteroids may be effective for treatment.
    c. Involvement of the vulva is common in women.
    d. Wide local excision may be necessary in refractory cases.
    e. Laser vaporization has been applied successfully to this condition.
23. A 35-year-old man presents with painful ulcerations in his mouth and on his penis, as well as blurred vision and a history of recurrent epididymitis. What is the likely diagnosis?
    a. Behçet disease

b. Oculocutaneous aphthous ulcer syndrome
c. Epidermolysis bullosa
d. Fabry disease
e. Pyoderma gangrenosum

24. Which of the following statements is FALSE concerning pyoderma gangrenosum?
    a. Pyoderma gangrenosum most likely has an autoimmune mechanism of pathogenesis.
    b. There is an association with collagen vascular disease.
    c. Corticosteroids may play a role in management.
    d. The presence of vacuolated keratinocytes in an inflammatory background is pathognomonic for this condition.
25. Which of the following cutaneous conditions has an association with borderline personality disorder?
    a. Factitial dermatitis
    b. "Innocent" traumatic dermatitis
    c. Sézary syndrome
    d. Munchausen syndrome by proxy
    e. Behçet disease
26. The most common organisms causing erysipelas are:
    a. dermatophytes.
    b. *Staphylococcus aureus.*
    c. *Streptococcus pyogenes.*
    d. *Escherichia coli.*
    e. *Pseudomonas* sp.
27. Which of the following statements is FALSE concerning Fournier gangrene?
    a. The mortality rate even with modern treatment may be greater than 15%.
    b. Most cases of Fournier gangrene are caused by *S. pyogenes.*
    c. Alcoholism is a significant risk factor for development of Fournier gangrene.
    d. In severe cases, debridement may need to extend into the chest wall.
    e. Fournier gangrene can be caused by a cutaneous, urethral, or perirectal source of infection.

28. An 18-year-old woman develops a pruritic rash over her thighs and buttocks after using a whirlpool spa. Her face and upper extremities are spared. What is the likely diagnosis?
    a. Candidal intertrigo
    b. Pseudomonal folliculitis
    c. Contact dermatitis
    d. Scabies infestation
    e. Herpes simplex
29. Which of the following conditions has an association with hyperhidrosis?
    a. Atopic dermatitis
    b. Trichomycosis axillaris
    c. Hidradenitis suppurativa
    d. Psoriasis
    e. Genital lichen planus
30. A patient being treated for tinea cruris has significant scrotal involvement. What alternative diagnosis does this suggest?
    a. Seborrheic dermatitis
    b. Erythrasma
    c. Cutaneous candidiasis
    d. Hidradenitis suppurativa
    e. Contact dermatitis
31. Which of the following is a treatment for scabies that is contraindicated in pediatric patients?
    a. Lindane
    b. Dapsone
    c. Permethrin
    d. Ivermectin
    e. Doxycycline
32. Which of the following statements concerning Bowen disease is FALSE?
    a. Bowen disease and squamous cell carcinoma in situ are the same condition.
    b. Bowen disease involving the glans penis is termed *erythroplasia of Queyrat.*
    c. Bowen disease may be treated with topical imiquimod.

d. Bowen disease is associated with human papillomavirus types 6 and 11.
e. Mohs microsurgery may play a role when tissue preservation is critical.

33. Which of the following statements concerning verrucous carcinoma is TRUE?
    a. Verrucous carcinoma has a high propensity to metastasize.
    b. Verrucous carcinoma should not be treated with primary radiotherapy because of the risk of anaplastic transformation.
    c. Verrucous carcinoma is an exceedingly rare malignancy of the genitalia.
    d. Verrucous carcinoma is associated with human papillomavirus types 16 and 18.
    e. Verrucous carcinoma may grow very rapidly and destroy local tissue.
34. What is the most common site of presentation for Kaposi sarcoma in immunocompetent individuals?
    a. Chest
    b. Face
    c. Lower extremities
    d. Genitalia
    e. Palms
35. The following malignancy has been found concurrently in lesions of pseudoepitheliomatous, keratotic, and micaceous balanitis:
    a. basal cell carcinoma
    b. cutaneous T-cell lymphoma
    c. squamous cell carcinoma
    d. verrucous carcinoma
    e. Kaposi sarcoma
36. Which of the following statements about extramammary Paget disease (EPD) is FALSE?
    a. EPD is an adenocarcinoma.
    b. EPD is associated with another underlying malignancy in more than 60% of cases.
    c. EPD has been associated with malignancies of the urethra and bladder.

d. EPD lesions show vacuolated Paget cells on histopathologic exam.
e. The vulva is the most common genital site involved in women.

37. Patients with cutaneous T-cell lymphoma who develop hematologic involvement are given the diagnosis of:
    a. lymphoid papulosis.
    b. mycosis fungoides.
    c. pagetoid reticulosis.
    d. Sézary syndrome.
    e. Fabry disease.
38. Which of the following conditions has the most in common histologically with pearly penile papules?
    a. Psoriasis
    b. Tuberous sclerosis
    c. Molluscum contagiosum
    d. Herpes simplex
39. The most effective treatment for Zoon balanitis is:
    a. topical 5-fluorouracil.
    b. topical corticosteroids.
    c. circumcision.
    d. laser therapy.
    e. topical calcineurin inhibitors.

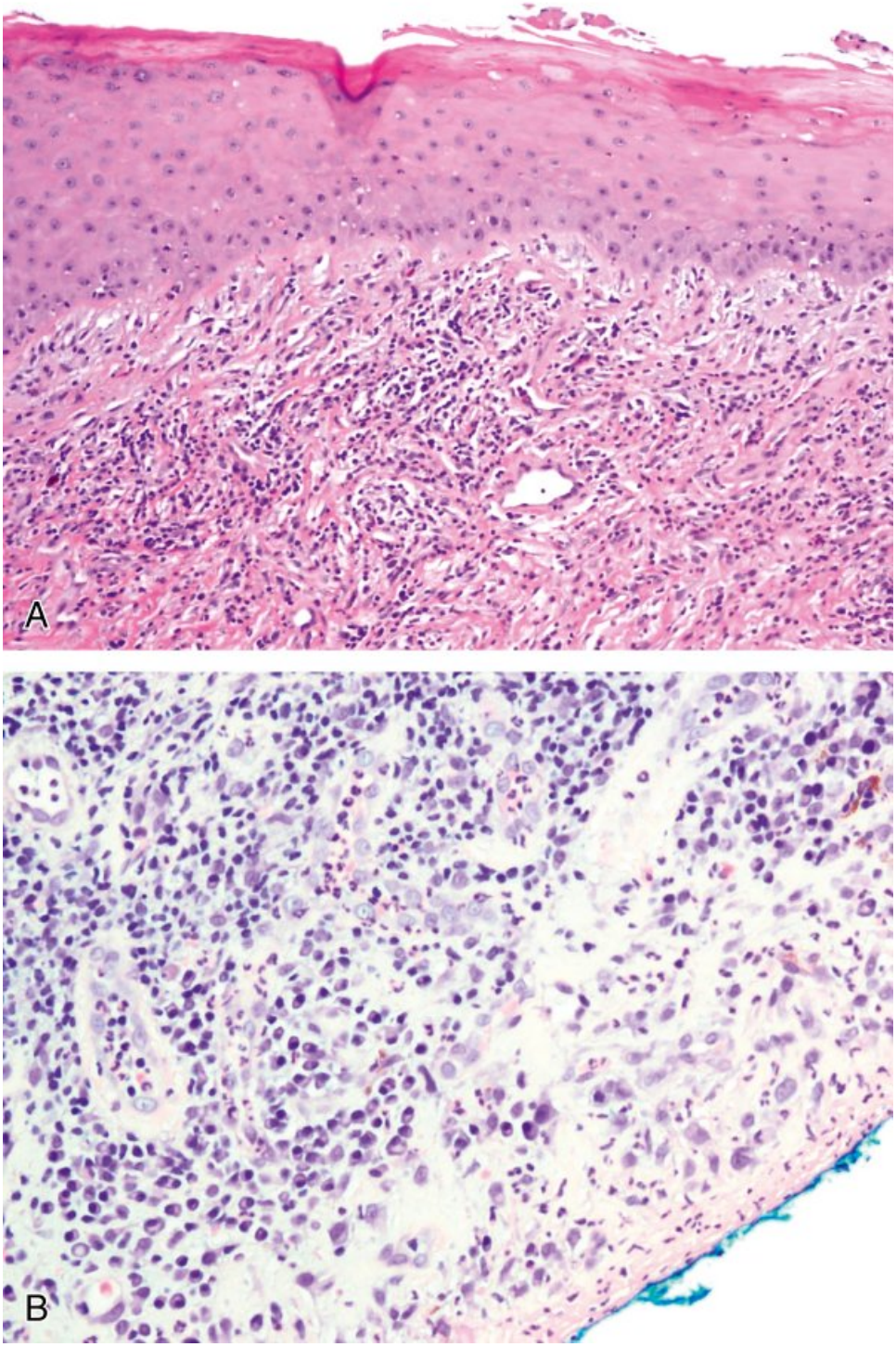


FIG. 59.1
From Bostwick DG, Cheng L: *Urologic surgical pathology*, 3rd ed, St. Louis, 2014, Saunders.

40. Skin tags are also termed:

a. fibrofolliculomas.
b. angiokeratomas.
c. hamartomas.
d. acrochordons.
e. dermatofibromas.

41. The Leser-Trélat syndrome refers to:
    a. the rapid progression of lichen planus associated with the HLA-B27 haplotype.
    b. an abrupt increase in the size and number of seborrheic keratoses, suggesting internal malignancy.
    c. the combination of hand, foot, and genital psoriasis.
    d. the development of brown macules on the genitalia, unrelated to sun exposure.
    e. the combination of oral and genital ulcers often seen in Behçet disease.
42. A 70-year-old uncircumcised man has noted an erythematous macular lesion on his glans at the corona. The pathology report of the biopsy (Fig. 59.1) reads plasma cell infiltrate consistent with Zoon balanitis. The next step in management is:
    a. ask the pathologist if these are CD4 cells consistent with exposure to HIV.
    b. laser fulguration.
    c. ask the pathologist if he or she looked for an associated squamous cell carcinoma.
    d. circumcision.
    e. observation.

Answers

1. **b. *Candida.*** To identify cutaneous fungi, such as dermatophytes and *Candida* species, periodic acid–Schiff (PAS) staining may be applied to scraped or touched skin specimens.
2. **d. 8 to 48 hours.** Oral glucocorticosteroids (GCS) are absorbed in the jejunum with peak plasma concentrations occurring in 30 to 90 minutes. Despite short plasma half-lives of 1 to 5 hours, the

duration of effect of GCS is between 8 and 48 hours, depending on the agent.

3. **a. A single morning dose.** For short-term (≤3 weeks) treatment of dermatologic conditions such as allergic contact dermatitis, a single morning dose of oral glucocorticosteroids is given to minimize suppression of the hypothalamic-pituitary-adrenal axis.
4. **d. The frequent application of emollients.** The condition described is atopic dermatitis (AD or eczema), which is associated with susceptibility to irritants and proteins, as well as the tendency to develop asthma and allergic rhinitis. Intense pruritus is the hallmark of AD, and controlling the patient's urge to scratch is critical for successful treatment. Removal of various "trigger factors" from the environment (such as chemicals, detergents, and household dust mites) may be beneficial in some cases. The mainstay of treatment for AD includes gentle cleaning with nonalkali soaps and the frequent use of emollients.
5. **b. Contact dermatitis.** Patch testing is a simple technique of exposing an area of skin to a variety of potential allergens in a grid template. Generally performed by dermatologists, patch testing can help confirm the diagnosis of allergic contact dermatitis and the allergen involved.
6. **d. Nickel sulfate.** In 2003, the North American Contact Dermatitis Group (NACDG) reported a long list of common allergens implicated in allergic contact dermatitis based on patch testing results. The most common sensitizing allergen identified was nickel sulfate, which is a common component of costume jewelry and belt buckles.
7. **d. Irritant contact dermatitis.** Irritant contact dermatitis results from a direct cytotoxic effect of an irritant chemical touching the skin and is responsible for approximately 80% of contact dermatitis cases. Occupational exposure is also common. Examples of offending agents include soaps, metal salts, acid- or alkali-containing compounds, and industrial solvents.
8. **c. Observation. Erythema multiforme (EM) minor is an acute, self-limited skin disease characterized by the abrupt onset of symmetrical fixed red papules that may evolve into target**

lesions. The majority of cases are precipitated by herpesvirus type I and II, with herpetic lesions usually preceding the development of target lesions by 10 to 14 days. Although continuous suppressive acyclovir may prevent EM episodes in patients with herpes infection, administration of the drug after development of target lesions is of no benefit. With observation alone, the natural history of EM minor is spontaneous resolution after several weeks without sequelae, although recurrences are common.

9. **c. Stevens-Johnson syndrome.** Stevens-Johnson syndrome (SJS) is a life-threatening severe allergic reaction with features similar to extensive skin burns. A vast array of inciting factors has been implicated in the development of SJS, with drug exposures being the most commonly identified. Nonsteroidal antiinflammatory agents are the most frequent offending agents, followed by sulfonamides, tetracycline, penicillin, doxycycline, and anticonvulsants.
10. **a. Topical 3% liquor carbonis detergens in 1% hydrocortisone cream.** Psoriasis is a papulosquamous disorder affecting up to 2% of the population with a relapsing and remitting course. **For genital psoriasis, the mainstay of therapy is the use of low-potency topical corticosteroid creams for short courses.** Photochemotherapy combining an ingested psoralen with ultraviolet radiation (PUVA) has been used extensively to treat psoriasis. However, a dose-dependent increase in the risk of genital squamous cell carcinoma has been associated with high-dose PUVA therapy for psoriasis elsewhere on the body. Genital shielding during PUVA therapy is strongly recommended; therefore, this modality is contraindicated for treating psoriatic lesions localized to genital skin.
11. **b. The human leukocyte antigen (HLA)-B27 haplotype. Reiter syndrome is a syndrome composed of urethritis, arthritis, ocular findings, oral ulcers, and skin lesions.** It is generally preceded by an episode of either urethritis *(Chlamydia, Gonococcus)* or gastrointestinal infection (*Yersinia, Salmonella, Shigella, Campylobacter, Neisseria,* or *Ureaplasma* species) and is

more common in HIV-positive patients. **There is a strong genetic association with the HLA-B27 haplotype.**

12. **c. Systemic corticosteroids can shorten the time to clearance of existing lesions from 29 to 18 weeks.** Although bothersome pruritus is common with lichen planus (LP), asymptomatic lesions on the genitalia do not require treatment. The primary modality of treatment for symptomatic lesions is topical corticosteroids, although for severe cases, systemic corticosteroids have been shown to shorten the time course to clearance of LP lesions from 29 to 18 weeks.
13. **d. Balanitis xerotica obliterans.** Lichen sclerosus (LS) is a scarring disorder, with a predilection for the external genitalia of both sexes, characterized by tissue pallor, loss of architecture, and hyperkeratosis. The late stage of this disease is called *balanitis xerotica obliterans*, which can involve the penile urethra and result in troublesome urethral stricture disease.
14. **a. Lichen sclerosus et atrophicus.** Despite the similarities in name, LS shares little in common with lichen planus and lichen nitidus other than pruritus and a predilection for the genital region. Another critical distinction is that LS has been associated with squamous cell carcinoma of the penis, particularly those variants not associated with human papillomavirus, and may represent a premalignant condition. Biopsy is worthwhile both to confirm the diagnosis and to exclude malignant change.
15. **b. Consult with neurology to alter his antiseizure medication regimen.** The association of epileptic seizures and a solitary painful genital lesion is suspicious for a diagnosis of Behçet disease (BD). Other causes for genital ulceration, however, including aphthous ulcers, syphilis, herpes simplex, and chancroid, must be considered before a diagnosis of BD is made. In this case, the patient's neurologic issues should take priority over treatment for his genital ulcer.
16. **b. HIV test.** Seborrheic dermatitis (SD) is a common skin disease characterized by the presence of sharply demarcated, pink-yellow to red-brown plaques with a flaky scale. Particularly in immunosuppressed individuals, SD may involve a significant

proportion of the body surface area. **Extensive and/or severe SD should raise concerns for possible underlying HIV infection.**

17. **b. The Asboe-Hansen sign.** The loss of epidermal cohesion seen in pemphigus vulgaris leads to the characteristic Asboe-Hansen sign: spreading of fluid under the adjacent normal-appearing skin away from the direction of pressure on a blister.
18. **e. Given enough time, even advanced cases of pemphigus generally resolve spontaneously without sequelae.** Severe cases of pemphigus vulgaris without appropriate treatment may be fatal because of the loss of the epidermal barrier function of large areas of affected skin. Treatment usually depends on systemic corticosteroids, although minimization of steroid dose is an important goal to limit side effects. The addition of immunosuppressive agents, such as azathioprine and cyclophosphamide, may be beneficial because of their corticosteroid-sparing effect.
19. **a. Dermatitis herpetiformis.** Dermatitis herpetiformis (DH) is a cutaneous manifestation of celiac disease and is generally associated with gluten sensitivity. Diagnosis can be confirmed by biopsy and direct immunofluorescence, which shows a granular pattern of IgA deposition at the basement membrane. Treatment includes the use of dapsone and a strict gluten-restricted diet.
20. **b. Pyoderma gangrenosum. Pyoderma gangrenosum (PG) is an ulcerative skin disease associated with systemic illnesses, including inflammatory bowel disease, arthritis, collagen vascular disease, and myeloproliferative disorders.** The classic morphologic presentation of PG is painful cutaneous and mucous membrane ulceration, often with extensive loss of tissue and a purulent base.
21. **c. Dapsone.** Characteristic clinical features of linear IgA bullous dermatosis (LABD) include vesicles and bullae arranged in a combination of circumferential and linear orientations. Treatment with either sulfapyridine or dapsone is usually effective in controlling LABD, and long-term spontaneous remission rates of 30% to 60% have been described.

22. **c. Involvement of the vulva is common in women.** Hailey-Hailey disease (HH) is an autosomal dominant blistering dermatosis that has a predilection for the intertriginous areas, including the groin and perianal region. Symptoms include an unfortunate combination of pruritus, pain, and a foul odor. Because heat and sweating exacerbate the condition, HH tends to worsen during the summer months. In women, disease in the inframammary folds is common, although vulvar disease is unusual. For disease resistant to medical therapy, wide excision and skin grafting have been effective, as have local ablative techniques such as dermabrasion and laser vaporization.
23. **a. Behçet disease. When oral and genital aphthous ulcers are coexistent, the clinician should consider the diagnosis of BD.** BD is a generalized relapsing and remitting ulcerative mucocutaneous disease that likely involves a genetic predisposition and an autoimmune mode of pathogenesis. Affected individuals may also suffer from epididymitis, thrombophlebitis, aneurysms, and gastrointestinal, neurologic, and arthritic problems.
24. **d. The presence of vacuolated keratinocytes in an inflammatory background is pathognomonic for this condition.** Pyoderma gangrenosum (PG) is an ulcerative skin disease associated with systemic illnesses, including inflammatory bowel disease, arthritis, collagen vascular disease, and myeloproliferative disorders. It most commonly affects women between the second and fifth decade of life and likely has an autoimmune pathogenesis, given its association with other autoimmune diseases. As is the case in Behçet disease, no specific diagnostic laboratory test or histopathologic feature is pathognomonic for PG, although a history of underlying systemic disease may raise suspicion.
25. **a. Factitial dermatitis.** Factitial dermatitis is a psychocutaneous disorder in which the individual self-inflicts cutaneous lesions, usually for an unconscious motive. An association between factitial dermatitis and borderline personality disorder appears to exist.
26. **c. *Streptococcus pyogenes.*** Erysipelas is a superficial bacterial skin infection limited to the dermis with lymphatic involvement. In contrast to the cutaneous lesion of cellulitis, erysipelas generally

has a raised and distinct border at the interface with normal skin. The causative organism is usually *S. pyogenes*.

27. **b. Most cases of Fournier gangrene are caused by *S. pyogenes.*** Fournier gangrene (FG) is a potentially life-threatening progressive infection of the perineum and genitalia. **In the genital region, most cases of FG are caused by mixed bacterial flora, which include gram-positive, gram-negative, and anaerobic bacteria.**
28. **b. Pseudomonal folliculitis.** Folliculitis is a common disorder characterized by perifollicular pustules on an erythematous base. It occurs most frequently in heavily hair-bearing areas such as the scalp, beard, axilla, groin, and buttocks and can be exacerbated by local trauma from shaving, rubbing, or clothing irritation. Folliculitis has also been associated with the use of contaminated hot tubs and swimming pools, with the offending organism usually *Pseudomonas aeruginosa.*
29. **b. Trichomycosis axillaris.** Trichomycosis axillaris (TA) is a superficial bacterial infection of axillary and pubic hair caused by *Corynebacterium*, which is associated with hyperhidrosis. Shaving can provide immediate improvement, and antibacterial soaps may prevent further infection. For pubic TA, clindamycin gel, bacitracin, and oral erythromycin have also proven effective.
30. **c. Cutaneous candidiasis. Tinea cruris is the term given to dermatophyte infection of the groin and genital area and is commonly known as "jock itch." The inner thighs and inguinal region are the most commonly affected areas, and the scrotum and penis are usually spared in men.** Significant scrotal involvement should raise suspicion for cutaneous candidiasis as an alternative diagnosis.
31. **a. Lindane.** As in the case of pediculosis pubis, the treatment of choice for scabies is 5% permethrin cream applied to the entire body overnight, with a second application 1 week later. An alternative scabicide, lindane, is not favored because of both central nervous system (CNS) toxicity in children and a rising rate of resistance among mites.
32. **d. Bowen disease is associated with human papillomavirus (HPV) types 6 and 11.** Bowen disease occurring on the mucosal

surfaces of the male genitalia is referred to as erythroplasia of Queyrat. In that location, coinfection with HPV type 8, 16, 39, and 51 has been identified. In contrast, the variant of squamous cell carcinoma termed "verrucous carcinoma" has been associated with HPV type 6 and 11 infection but not with the more classically oncogenic type 16 and 18.

33. **b. Verrucous carcinoma (VC) should not be treated with primary radiotherapy because of the risk of anaplastic transformation. VC is a slow-growing, locally aggressive, exophytic, low-grade variant of squamous cell carcinoma that has little metastatic potential.** It most commonly occurs in uncircumcised men on the glans or prepuce, although similar lesions can be found on the vulva, vagina cervix, or anus. Treatment is preferably by local excision. Primary radiotherapy is relatively contraindicated because of the potential for anaplastic transformation with a subsequent increase in metastatic potential.
34. **c. Lower extremities.** Kaposi sarcoma (KS) in immunocompetent individuals presents as slowly growing, blue-red pigmented macules on the lower extremities. Although oral and gastrointestinal lesions may occur, the genitalia are seldom involved. This is in contrast to the case with acquired immunodeficiency syndrome (AIDS), in which a solitary genital lesion may be the first manifestation of KS. The clinical features of KS in AIDS patients are diverse, ranging from a single lesion to disseminated cutaneous and visceral disease.
35. **d. Verrucous carcinoma.** Pseudoepitheliomatous, keratotic, and micaceous balanitis (PEKMB) is a rare entity characterized by the development of a thick, hyperkeratotic plaque on the glans penis of older men. There remains controversy as to whether PEKMB is a premalignant condition. PEKMB was originally thought to be a purely benign process, although several case reports have documented the presence of concurrent verrucous carcinoma associated with this lesion.
36. **b. EPD is associated with another underlying malignancy in more than 60% of cases. Extramammary Paget disease (EPD) is an uncommon intraepithelial adenocarcinoma of sites**

bearing apocrine glands. There is an important association between EPD and another underlying malignancy in 10% to 30% of cases. In the male, associations of urethral, bladder, rectal, and apocrine malignancies with EPD have been described.

37. **d. Sézary syndrome.** Cutaneous T-cell lymphoma (CTCL) represents a group of related neoplasms derived from T cells that home to the skin. CTCL generally presents with pruritus, which must be differentiated from a variety of benign dermatoses, including psoriasis, eczema, superficial fungal infections, and drug reactions. Patients may subsequently develop hematologic involvement (termed Sézary syndrome) and cutaneous plaques, erosions, ulcers, or frank skin tumors.
38. **b. Tuberous sclerosis. Pearly penile papules (PPP) are white, dome-shaped, closely spaced, small papules located on the glans penis. Histologically, these lesions are angiofibromas similar to the lesions seen on the face in tuberous sclerosis.**
39. **c. Circumcision.** Zoon balanitis, also called plasma cell balanitis, occurs in uncircumcised men from the third decade onward. Squamous cell carcinoma and extramammary Paget disease should be excluded, often by biopsy. Circumcision appears to be proof against development of the disease and can be performed to cure the majority of cases. For patients averse to circumcision, topical corticosteroids may provide symptomatic relief, and laser therapy may also have a role.
40. **d. Acrochordons.** Skin tags (acrochordons, fibroepithelial polyps) are soft, skin-colored, pedunculated lesions that can be present anywhere on the body. It is important to distinguish these lesions from the hamartomatous skin lesions (multiple fibrofolliculomas) associated with Birt-Hogg-Dubé syndrome, which are histologically distinct from common skin tags.
41. **b. An abrupt increase in the size and number of seborrheic keratoses, suggesting internal malignancy. The presence of brown macules unrelated to sun exposure suggests a diagnosis of SK.** This condition may commonly involve the genitalia but generally spares the mucous membranes, palms, and soles of the feet. An abrupt increase in the size and number of multiple SK has

been termed Leser-Trélat syndrome and has been implicated as a cutaneous marker of internal malignancy. The HLA-B27 haplotype is associated with Reiter syndrome, not rapidly progressive lichen planus.

42. **d. Circumcision.** The patient has Zoon balanitis. The biopsy shows reactive epithelial changes with a plasma cell infiltrate into the dermis. Although the disease can be treated with immune modulators, these patients respond best to circumcision.

Chapter Review

1. Topical application of steroids may result in systemic absorption resulting in significant side effects.
2. The majority of cases of erythema multiforme are precipitated by herpesvirus type I and II, with the herpetic lesions usually preceding the development of the target lesions by 2 weeks.
3. The major form of erythema multiforme is called Stevens-Johnson syndrome, which has a protracted course of 4 to 6 weeks and may have a mortality approaching 30%. Nonsteroidal antiinflammatory agents, sulfonamides, tetracyclines, penicillin, doxycycline, and anticonvulsants are the most common offending agents.
4. Reiter syndrome comprises urethritis, arthritis, ocular findings (conjunctivitis), oral ulcers, and skin lesions. It is generally preceded by an episode of urethritis or a gastrointestinal infection. It is more common in HIV-positive patients. It is associated with HLA-B27 haplotype.
5. The triad of clinical features in Behçet syndrome consists of mucocutaneous lesions of the oral cavity, genitalia, and uveitis. The ulcers are painful.
6. Hidradenitis suppurativa is an epithelial disorder of hair follicles that occurs in the apocrine gland–bearing skin, which results in a marked inflammatory response with formation of abscesses and sinus tracks. Bacterial infection does not appear to be the primary initiator. Rupture of the follicular contents—bacteria and keratin—into the dermis causes a marked inflammatory response.

7. Ecthyma gangrenosum is a result of pseudomonal septicemia and may result in gangrenous ulcers.
8. Dermatophytes are fungi of three genera: *Trichophyton, Microsporum,* and *Epidermophyton.* Tinea cruris, an infection of the groin and genital skin, may be caused by one of these fungi. Postinflammatory hyperpigmentation occurs with this disease and may not indicate an active infection.
9. Bowenoid papulosis consists of multiple small erythematous papules and is associated with HPV type 16.
10. Angiokeratomas of Fordyce are vascular ectasias of dermal blood vessels. They are 1- to 2-mm red or purple papules and may be the source of troublesome scrotal bleeding.
11. CIS of the penis rarely progresses to invasive disease.
12. Whether Kaposi sarcoma is a neoplastic or hyperplastic process is debated. When it is associated with organ transplantation, alteration of the immunosuppressive regimen may result in resolution.
13. Epidermoid cysts are commonly found in the scrotum.
14. Extensive and/or severe seborrheic dermatitis should raise concerns for possible underlying HIV infection.
15. Pyoderma gangrenosum is an ulcerative skin disease associated with systemic illnesses, including inflammatory bowel disease, arthritis, collagen vascular disease, and myeloproliferative disorders.
16. In tinea cruris, the inner thighs and inguinal region are the most commonly affected areas, and the scrotum and penis are usually spared in men.
17. Extramammary Paget disease (EPD) is an uncommon intraepithelial adenocarcinoma of sites bearing apocrine glands. There is an important association between EPD and another underlying malignancy in 10% to 30% of cases.